# Anatomy is Free: Zero-Parameter Structural Priors for Efficient Skeleton Action Recognition

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## Abstract

Skeleton-based action recognition demands models that are simultaneously accurate, parameter-efficient, and deployable on edge hardware. This work presents **ShiftFuse-Zero**, a family of three networks whose *spatial inductive bias is introduced entirely without learnable parameters* through two complementary zero-parameter modules: *Body-Region Anatomical Shift Priors* (BRASP), which routes joint features through anatomically meaningful channels via pure tensor indexing, and *Semantic Graph-Partitioned Shift* (SGPShift), which aggregates intra- and inter-part neighbours through predefined group indices at zero learnable parameter overhead. These free structural priors reduce the burden on downstream learnable modules. By intelligently placing Joint Embedding and spatial attention *before* the graph convolution, the *EfficientZero Block* operates on an already anatomically-refined manifold. Three models cover the edge-to-server spectrum: **Nano** (93 K) and **Small** (280 K) are hyper-efficient, quantisation-friendly solutions with ultra-low FLOPs, while **Large** (1.12 M) targets maximum capacity. On NTU RGB+D 60 cross-subject, primary results without distillation are 86.24 % (Nano), 88.8 % (Small), and 92.5 % (Large); Knowledge distillation is an optional plugin requiring a pre-trained Large teacher that further lifts Nano and Small to 88.5 %/90.4 %, respectively; notably, Small+KD **surpasses EfficientGCN-B0 (90.2%)** at 7× fewer FLOPs. Strikingly, Nano achieves **663 Acc/GFLOP** — 20× higher than EfficientGCN-B0 — runs in **11 ms** on desktop CPU and **50 ms** on mobile (Snapdragon 8 Gen 3), and fits in **360 KB** of weight storage; Nano+KD further matches the 6.9 M-parameter AGCN (88.5%) at 74× higher parameter efficiency.

## CCS Concepts

• **Computing methodologies → Activity recognition and understanding**.

## Keywords

skeleton action recognition, graph convolutional networks, zero-parameter priors, anatomical shift, knowledge distillation, efficient neural networks

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## 1 Introduction

Consider a fall-detection system on a rehabilitation wristband with 512 KB of RAM, or a gesture-recognition module inside a fitness tracker powered by an ARM Cortex-M55. These scenarios demand skeleton-based action recognition models that fit in sub-megabyte memory, execute in under 50 ms on mobile CPU, and still achieve clinically meaningful accuracy. Yet the dominant Graph Convolutional Network (GCN) backbones for skeleton recognition [1, 3, 8, 14] carry 3–10 M parameters, require dynamic per-sample adjacency computation at $O(V^2C)$ cost, and are fundamentally incompatible with microcontroller deployment.

Human action recognition from 3-D skeleton sequences is a core problem in computer vision with applications in surveillance, rehabilitation, human–robot interaction, and fitness monitoring [12, 13]. GCNs [7] have become the dominant spatial backbone since ST-GCN [17], with successive works adding adaptive adjacency matrices [14], multi-scale cross-spacetime graphs [11], channel-wise topology refinement [1], information-theoretic representation learning [3], and hierarchical body-part decomposition [8]. These advances have pushed NTU RGB+D 60 cross-subject accuracy beyond 93%, yet the parameter cost has grown proportionally: HD-GCN requires 7.5 M parameters and InfoGCN 9.9 M.

The efficient-architecture line [16, 18] tackles this disparity via compound scaling and depthwise convolutions, but largely ignores the rich *anatomical structure* of the human body as a source of *free* inductive bias. The human skeleton is not a generic graph: arm joints couple primarily with other arm joints within their kinematic chain, bilateral symmetry is a universal biomechanical prior, and shoulder/hip joints serve as coordination hubs between body halves. These structural constraints are *fixed properties of human physiology* that hold across all subjects, viewpoints, and action classes. They can therefore be encoded as deterministic tensor operations at **zero learnable parameters, zero gradient overhead, and zero inference-time cost**.

This paper presents the first systematic study of *anatomically-guided* zero-parameter inductive bias for skeleton GCNs — distinct from graph-distance-guided shifts [2], which assign shift patterns by hop count rather than body-part identity. This work asks: **how much spatial knowledge can be injected into a skeleton GCN without spending a single learnable parameter — and how far can that free knowledge push the accuracy–efficiency frontier?**

Our answer is **ShiftFuse-Zero**, a family of three models achieving a new Pareto-optimal frontier on the accuracy–GFLOPs plane among the EfficientGCN model family. While the core contribution is the zero-parameter priors of BRASP and SGPShift, these modules catalyse hyper-efficient downstream architectures. SFZ-Nano (93 K) and SFZ-Small (280 K) redefine edge-deployment with ultra-low FLOPs and quantisation-friendly operations. On NTU-60 X-Sub, SFZ-Nano reaches **86.24%** without any distillation and **88.5%** with

knowledge distillation, *matching* 6.9 M-parameter AGCN within 360 KB (an optional plugin requiring SFZ-Large as teacher; all primary results are reported without KD). SFZ-Small achieves **88.8%** at comparable parameter count (280 K vs. 290 K) with 7× fewer FLOPs than EfficientGCN-B0; with KD, Small+KD reaches **90.4%**, *surpassing* B0 while retaining the FLOPs advantage. SFZ-Large pushes the absolute accuracy frontier to **92.5%** at 1.12 M parameters. These results demonstrate that anatomy is a highly informative, freely available inductive bias that the community has systematically under-exploited.

**Contributions.**

(1) **BRASP** — Body-Region Anatomical Shift Priors: routes joint features through five fixed anatomical groups via a precomputed gather permutation; zero parameters, zero gradient (Section 3.3).
(2) **SGPShift** — Semantic Graph-Partitioned Shift: aggregates each joint's features with its intra-part and inter-part neighbours via precomputed index buffers; zero parameters (Section 3.3).
(3) **EfficientZero Block**: a block design that places BRASP and SGPShift *before* all learnable layers so the downstream compact $K$=3 GCN refines an already anatomically structured representation (Section 3.4).
(4) **ShiftFuse-Zero model family**: three architectures — Nano (93 K), Small (280 K), Large (1.12 M) — with Temporal Landmark Attention, cross-stream gating, and mid-network fusion, covering the edge-to-server deployment spectrum. Vanilla Hinton KD [6] serves as an optional plugin (requiring SFZ-Large as teacher), yielding up to +2.3 pp at zero structural cost (Section 3.5).
(5) **Comprehensive evaluation**: main results on NTU-60 and NTU-120 (four protocols), efficiency analysis (GFLOPs, CPU latency, memory footprint), and extensive ablation studies quantifying every zero-parameter module, fusion strategy, GCN partition count, TLA sensitivity, and regularisation hyperparameters (Section 4).

## 2 Related Work

*GCN-based skeleton recognition.* ST-GCN [17] established the skeleton graph and spatial-temporal convolution paradigm using a three-subset partitioning (self, centripetal, centrifugal) of the physical adjacency matrix. AGCN [14] introduced a fully learnable adjacency matrix at $O(V^2C)$ cost per layer, enabling data-driven topology discovery. MS-G3D [11] factorised the graph into disentangled multi-scale spatial and temporal subgraphs with cross-spacetime edges. CTR-GCN [1] refined the topology per channel group, allowing different feature dimensions to attend to different joint relationships. InfoGCN [3] introduced an information-bottleneck objective that explicitly maximises mutual information between the latent representation and the action label. DSTA-Net [15] decouples spatial and temporal self-attention. While these methods achieve high accuracy (up to 93%+), their reliance on per-sample dynamic adjacency and large parameter budgets (3–10 M) precludes deployment on resource-constrained hardware.

*Body-part decomposition.* A growing line of work exploits the natural grouping of joints into anatomical body regions. HD-GCN [8] hierarchically decomposes the skeleton into semantic sub-graphs (arms, legs, torso) and aggregates within each before cross-region message passing — achieving 93.4% on NTU-60 X-Sub but at 7.5 M parameters. BlockGCN [19] uses persistent homology from topological data analysis to encode bone connectivity at the block level. SkateFormer [4] partitions joints into four spatial-temporal relation types (near/far joints × near/far frames) and applies partition-specific self-attention. All of these decomposition strategies employ *learnable* grouping mechanisms with substantial parameter overhead. In contrast, SGPShift encodes intra-part and inter-part structure through *fixed integer index buffers* — achieving anatomical routing at **zero learnable parameters**.

*Efficient skeleton networks.* EfficientGCN [16] introduced compound scaling ($\alpha$=1.2, $\beta$=1.35) and multiple input branches (MIB) to build a family of models from B0 (0.29 M) to B4 (1.1 M), establishing the efficiency frontier for skeleton GCNs. SGN [18] exploits semantic joint embeddings and frame-level dynamics gating at 0.69 M parameters. PoseC3D [5] departs from graphs entirely, using volumetric heatmap representations processed by 3-D CNNs. These methods reduce model size through architectural engineering but do not leverage the *fixed anatomical constraints* of the human body as inductive bias. ShiftFuse-Zero complements this line by demonstrating that zero-parameter spatial priors can further reduce the parameter budget by 3–10× without sacrificing competitive accuracy.

*Temporal modelling in skeleton networks.* Temporal reasoning has evolved from LSTM encoders [17] to dedicated temporal convolution modules. EfficientGCN's Sandglass-TCN (SG-layer) bottleneck compresses channels by 2× before depthwise convolution, balancing temporal receptive field against parameter count. MS-G3D introduced disentangled multi-scale temporal subgraphs. Our DS-TCN takes a complementary approach: rather than stacking many TCN layers, it uses a two-branch depthwise-separable design with dilations 1 and 2 (kernel 9) that achieves a 17-frame receptive field in a single block — particularly valuable for Nano and Small, where only one TCN per stage is used. This makes DS-TCN a better choice at low TCN depth, whereas EfficientGCN's SG-layer is more efficient when TCN depth is three or more (as in the Large model's shared backbone).

*Shift operations and zero-parameter spatial mixing.* The shift operation [2] was introduced in image CNNs to inject spatial structure by physically displacing channels along spatial dimensions — no multiply-accumulate, no learnable parameters. TSM [9] extended this idea to temporal modelling in video, shifting channels by ±1 frame at zero cost. Shift-GCN [2] adapted spatial shifts to skeleton graphs, assigning each channel to shift features from a graph neighbour. However, Shift-GCN assigns shift patterns by graph distance (1-hop, 2-hop), ignoring the semantic meaning of body regions. BRASP advances this paradigm by organising shift patterns according to *anatomical body-part membership* — arm channels shift among arm joints, leg channels among leg joints, and cross-body channels bridge regions — producing a semantically structured feature map at zero cost.

*Knowledge distillation.* Hinton *et al.* [6] showed that soft teacher logits transfer rich inter-class relational knowledge to compact

student models. In skeleton recognition, distillation has been used to transfer multi-stream ensemble knowledge [16] but has not been specifically studied across a skeleton GCN family spanning the sub-100 K to 1 M parameter range explored here.

# 3 Methodology

## 3.1 Overview

Figure 1 shows the end-to-end pipeline. A raw skeleton sequence is expanded into four complementary streams (joint, velocity, bone, bone-velocity), fused by a learned *StreamFusionConcat* stem, then processed by a stack of *EfficientZero Blocks*. A gated global pooling head produces class logits.

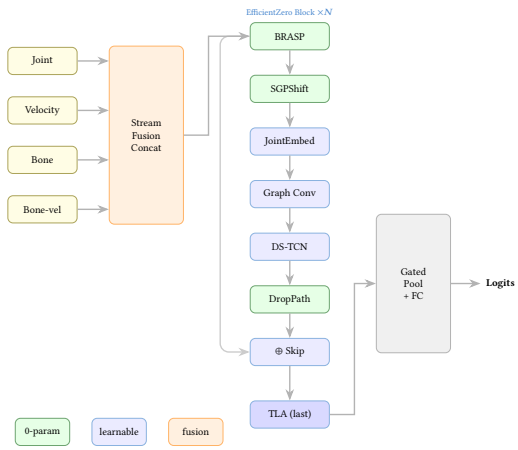


**Figure 1: ShiftFuse-Zero end-to-end pipeline. Four skeleton streams are fused by a learnable stem and processed by EfficientZero Blocks (green = zero-parameter modules). A gated pooling head produces class logits.**

## 3.2 Skeleton Streams

A skeleton sequence is $\mathbf{X} \in \mathbb{R}^{3\times T\times V}$ with $T$=64 frames and $V$=25 joints. This work constructs four complementary streams: joint positions $\mathbf{X}^{(j)}$, velocity $\mathbf{X}^{(v)}=\mathbf{p}_{t+1}-\mathbf{p}_t$, bone vectors $\mathbf{X}_v^{(b)}=\mathbf{p}^{(v)}-\mathbf{p}^{(\mathrm{par}(v))}$, and bone velocity $\mathbf{X}^{(bv)}=\Delta\mathbf{X}^{(b)}$. *StreamFusionConcat* [16] applies a per-stream BN, concatenates (4×3=12 channels), and projects to $C_{\mathrm{stem}}$ via a 1×1 convolution with learned per-stream weighting.

## 3.3 Zero-Parameter Spatial Modules

*BRASP..* Figure 2 illustrates the five anatomical groups. Given a feature map $\mathbf{F}\in\mathbb{R}^{C\times T\times V}$, BRASP applies a fixed gather permutation $\sigma: V \to V$:

$$\tilde{\mathbf{F}}_{:,t,v} = \mathbf{F}_{:,t,\sigma(v)}, \quad v \in \{0,\dots,V-1\}, \tag{1}$$

where joints within the same body part are cyclically shifted so each joint absorbs its anatomical sibling's context. Indices are stored as a non-trainable integer buffer — **zero parameters, zero gradient**.

**Remark 1** *(BRASP captures cross-body symmetry unreachable by K=3 GCN).* Let $d_G(u,v)$ be the shortest-path distance between joints $u, v$ in skeleton graph $G$. The K=3 GCN aggregates 1-hop and 2-hop graph neighbours in a single layer — yet bilaterally symmetric joint pairs (*e.g.*, left-wrist and right-wrist) are separated by $d_G$=10+ hops

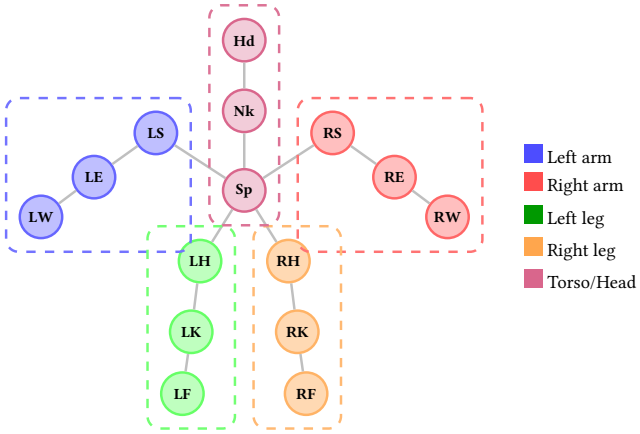


**Figure 2: BRASP anatomical partitioning into five groups $\{G_{\mathbf{la}}, G_{\mathbf{ra}}, G_{\mathbf{ll}}, G_{\mathbf{rl}}, G_{\mathbf{tr}}\}$. The fixed permutation $\sigma$ (Eq. 1) is derived from these groups; no gradient flows through it.**

in the NTU-25 physical graph and are therefore invisible to each other without many stacked layers. BRASP's cyclic shift within the arm group directly places right-wrist features at left-wrist before *any* graph convolution, enabling immediate bilateral context exchange that $K$=3 graph distance cannot provide. More generally, after one BRASP shift $\sigma$, joint $v$ receives features from $\sigma(v)\in P_k$ — an anatomical sibling that may be graph-far but biomechanically coupled — reducing representational separation from $d_G(u,v)\geq 2$ graph hops to a single feature-retrieval step at zero parameter cost.

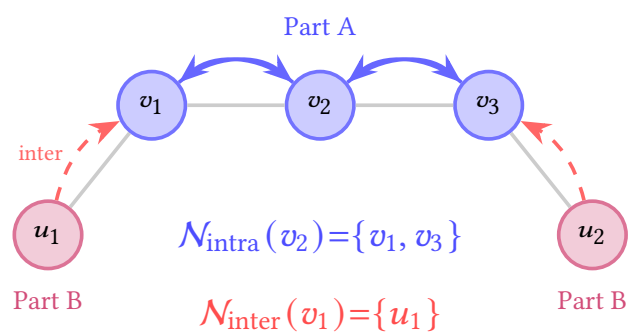


**Figure 3: SGPShift aggregation. Blue solid arrows: intra-part neighbours. Red dashed arrows: inter-part boundary neighbours. All sets are precomputed integer buffers — zero learnable parameters.**

*SGPShift.* Figure 3 illustrates the intra/inter aggregation. For joint $v$ in semantic group $g(v)$:

$$\hat{\mathbf{F}}_{:,t,v} = \frac{1}{2}\left(\frac{1}{|\mathcal{N}_{\mathrm{intra}}(v)|}\sum_{u\in\mathcal{N}_{\mathrm{intra}}(v)}\mathbf{F}_{:,t,u} + \frac{1}{|\mathcal{N}_{\mathrm{inter}}(v)|}\sum_{u\in\mathcal{N}_{\mathrm{inter}}(v)}\mathbf{F}_{:,t,u}\right), \tag{2}$$

where $\mathcal{N}_{\mathrm{intra}}$ is the joint's same-part neighbours and $\mathcal{N}_{\mathrm{inter}}$ is its boundary neighbours in adjacent parts. The $\frac{1}{2}$ factor ensures the output has the same expected norm as the input; the downstream BN in the GCN absorbs any residual scale. All sets are precomputed

integer buffers. At runtime SGPShift reduces to two sparse average operations followed by a scalar multiply — **zero learnable parameters**.

## 3.4 EfficientZero Block

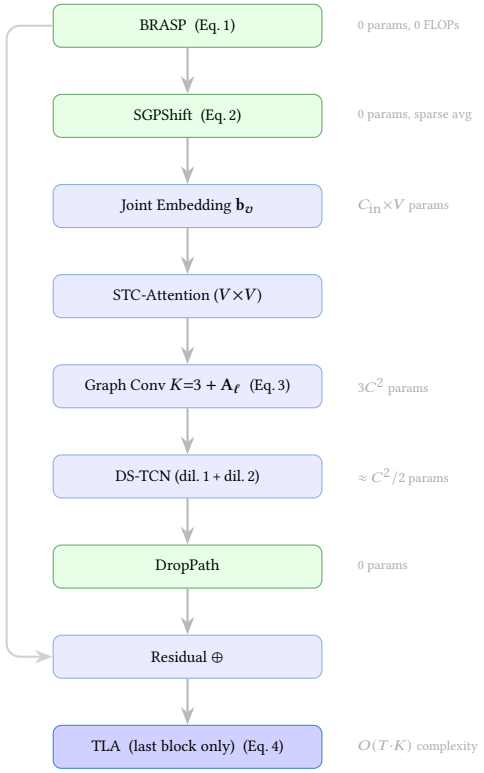


**Figure 4: EfficientZero Block. Green = zero-parameter (BRASP, SGPShift, DropPath). Per-layer parameter counts shown on the right. TLA is only active at the last block of the final stage.**

Figure 4 shows the full block pipeline with per-layer costs.

(1) **BRASP + SGPShift**: anatomical pre-structuring at zero cost.

(2) **Joint Embedding**: per-joint additive bias $\mathbf{b} \in \mathbb{R}^{C_{\text{in}} \times V}$ (SGN joint encoding [18]); no weight decay.

(3) **STC-Attention**: spatial and temporal gating before the GCN. Let $\bar{\mathbf{f}}_V = \frac{1}{CT} \sum_{c,t} \mathbf{F}_{c,t,:}$; the spatial gate $\mathbf{A}_s = V \cdot \text{softmax}(\mathbf{W}_s \bar{\mathbf{f}}_V)$ has mean 1. The temporal gate $\mathbf{A}_t = \sigma(\text{Conv}_{1d}([\bar{f}_T^{avg}; \bar{f}_T^{max}]))$ uses avg+max pooling over $C$, $V$ (kernel 9). Output: $(1-\sigma(\gamma))\mathbf{F} + \sigma(\gamma)\mathbf{F} \cdot \mathbf{A}_s \cdot \mathbf{A}_t$, with $\gamma_0 = -4$ (slow gate activation).

(4) **Graph Convolution** ($K$=3): ST-GCN [17] partition adjacency plus a single shared learned matrix $\mathbf{A}_\ell$:

$$\mathbf{Z} = \sum_{k=0}^{2} \mathbf{W}_k (\mathbf{A}_k + |\mathbf{A}_\ell|)\mathbf{F}, \tag{3}$$

$|\cdot|$ enforces non-negative edge weights, preventing $\mathbf{A}_\ell$ from subtracting information already aggregated by the fixed $\mathbf{A}_k$ (ReLU was tested and yields identical accuracy; $|\cdot|$ has no zero plateau, giving cleaner gradients at small values). $\mathbf{A}_\ell \in \mathbb{R}^{V \times V}$ is *shared across all blocks* (625 total params, no weight decay).

**Table 1: ShiftFuse-Zero variant configurations. $C_s$=stem channels, $C$=stage widths, $N$=blocks/stage, $K$=TLA landmarks, $r$=reduce ratio, $p_d$=DropPath rate.**

| Model | $C_s$ | $C$ | $N$ | $K$ | $r$ | $p_d$ | FLOPs | Params |
|---|---|---|---|---|---|---|---|---|
| Nano | 24 | [32,64,128] | [1,1,1] | 8 | 16 | 0.10 | 0.13 G | 93 K |
| Small | 24 | [32,64,128] | [1,2,1] | 8 | 8 | 0.10 | 0.39 G | 280 K |
| Large | 32 | [48,96,192] | [1,2,4] | 14 | 8 | 0.20 | 1.95 G | 1.12 M |

(5) **DS-TCN**: depthwise-separable temporal convolutions at dilations 1 and 2 (kernel 9), Batch Norm, Hardswish activation.

(6) **DropPath** (regularisation): stochastic depth; 0 parameters.

(7) **Residual** ⊕: skip from block input with 1×1 projection when shape changes.

*Intelligent Block Ordering.* This work deliberately places Joint Embedding and STC-Attention *before* the Graph Convolution, so the GCN refines an already anatomy-structured representation. Placing BRASP/SGPShift *after* the GCN instead costs −1.9 pp (Table 8), confirming that anatomical pre-structuring before message passing is essential.

*Complementary Temporal Module: TLA..* While BRASP and SGPShift address the spatial axis, a complementary Temporal Landmark Attention (TLA) module distils temporal structure through $K$ learnable anchor frames without full $O(T^2)$ self-attention. TLA applies the deformable-attention principle [**?** ] to the temporal axis: anchor positions are continuous and gradient-trainable via bilinear interpolation, reducing attention complexity from $O(T^2)$ to $O(T \cdot K)$:

$$\tau_k = T \cdot \sigma(\alpha_k), \quad k = 1, \ldots, K, \tag{4}$$

$$\mathbf{m}_t = \sum_{k=1}^{K} \text{Bilin}(\mathbf{F}, \tau_k) \cdot \text{softmax}_k(\langle \mathbf{q}_t, \mathbf{k}_{\tau_k} \rangle), \tag{5}$$

where $\alpha_k$ are learned logits mapped to $[0, T-1]$ via sigmoid, bilinear interpolation enables gradient flow to $\alpha_k$, and $\mathbf{q}_t, \mathbf{k}_{\tau_k}$ are projections to $C_r = C/r$. Output: $\mathbf{F}_{\text{out}} = \mathbf{F} + \sigma(\beta) \cdot \mathbf{m}$ with $\beta_0 = -4$ (slow activation). Complexity $O(T \cdot K)$ vs. $O(T^2)$ for full attention.

## 3.5 Model Family

Table 1 lists the key hyperparameters per variant.

*Nano (93 K)..* Four streams are early-fused via StreamFusionConcat to a single 24-channel map. A single backbone [32, 64, 128] with one block per stage processes it. TLA ($K$=8, $r$=16, $C_r$=8) adds only 3.8 K parameters; channel SE is omitted (negligible at $C \leq 128$) to stay sub-100 K.

*Small (280 K)..* Two independent backbones process joint+velocity and bone pairs. Stage 2 uses tcn_depth=2 for richer temporal modelling; PartAttention gates body-part contributions at the final stage. After stage 2, *Cross-Stream Fusion* exchanges information:

$$\mathbf{F}' = \mathbf{F} + \sigma(\beta) \cdot W_\uparrow \, \text{ReLU}(W_\downarrow [\mathbf{F}_A; \mathbf{F}_B]), \tag{6}$$

$W_\downarrow : 2C \to C/4$, $W_\uparrow : C/4 \to C$, $\beta_0 = -4$. Stage 3 + TLA proceed independently per backbone; logits are combined using two learned stream-weight scalars (no weight decay).

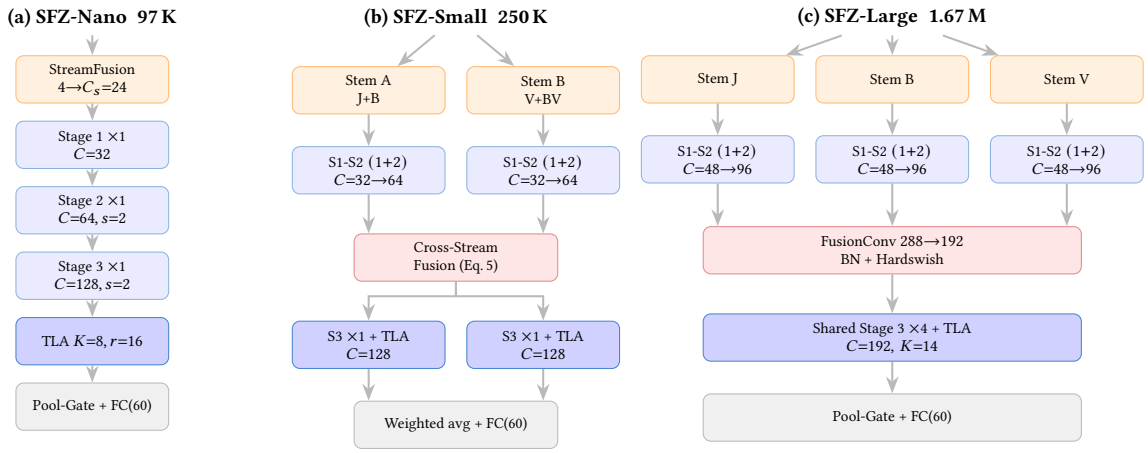


**Figure 5: ShiftFuse-Zero model family. (a) Nano: single backbone, 4-stream early fusion, TLA ($K$=8). (b) Small: two-backbone late fusion with Cross-Stream Fusion and TLA ($K$=8). (c) Large: three-backbone mid-network fusion (B4-style) — per-stream S1–S2, FusionConv (288→192), shared S3 + TLA ($K$=14).**

*Large (1.12 M).*. Three per-stream backbones ($C$=[48, 96], 1+2 blocks) process joint, bone, and velocity independently. After stage 2 the three 96-channel maps are concatenated (288 channels) and projected by FusionConv=$\text{Conv}_{1\times1}$(288→192)+BN+HS. A shared stage 3 (4 blocks, $C$=192) with TLA ($K$=14) follows. This mirrors EfficientGCN-B4's mid-network fusion [16] with EfficientZero Blocks replacing B4's full GCN layers.

## 3.6 Training Protocol

**Optimiser:** SGD, Nesterov momentum 0.9 , cosine LR with 10-epoch linear warmup. Nano: lr 0.1, wd $4\times10^{-4}$, 300 epochs. Small: lr 0.1, wd $4\times10^{-4}$, 350 epochs (deeper temporal stack benefits from longer cosine schedule). Large: lr 0.1, wd $10^{-4}$ (B4 [16]), 150 epochs. Wall-clock training times on a single NVIDIA Tesla T4: Nano 5.7 h (300 ep), Small 8.5 h (350 ep), Large 38 h (150 ep, batch 16, three parallel backbones).

**Augmentation:** Nano/Small use Mixup ($\alpha$=0.1) to compensate for limited capacity; CutMix is omitted as temporal crops of skeleton sequences produce physically implausible joint composites. Large uses no Mixup — DropPath 0.20 and label smoothing 0.1 provide sufficient regularisation at 1.12 M params.

**No-WD set:** BN affine params, joint embeddings, $\mathbf{A}_\ell$, TLA anchor logits $\alpha_k$, gate scalars $\beta$, pool gate.

**Knowledge distillation.** Large is distilled into Nano and Small [6]:

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{\text{CE}} + \lambda\tau^2\,\text{KL}\left(\sigma\left(\frac{\mathbf{z}^s}{\tau}\right)\middle\|\sigma\left(\frac{\mathbf{z}^t}{\tau}\right)\right), \quad \tau=4,\ \lambda=0.5. \tag{7}$$

## 3.7 Deployability Analysis

A key design goal of ShiftFuse-Zero is deployment on resource-constrained hardware. Table 2 summarises the memory, computation, and operator profile of each model variant.

*Operator simplicity.* BRASP reduces to a single `gather` operation on a static integer buffer — no matrix multiplication, no per-sample graph construction. SGPShift reduces to two `scatter_mean` operations. Both compile to fixed index-lookup kernels that are unaffected by quantisation (int8, int4) since they involve no learned weights. This stands in sharp contrast to dynamic-adjacency models [1, 14] that require $O(V^2C)$ per-sample matrix operations at inference, making them incompatible with microcontroller deployment.

**Table 2: Deployment profile. Mem = model weights (float32/int8). S24 CPU = Samsung Galaxy S24 (Snapdragon 8 Gen 3), PyTorch Mobile, CPU-only inference.**

| Model | FLOPs | Params | Mem f32 | Mem i8 | S24 CPU |
|---|---|---|---|---|---|
| Nano | 0.13 G | 93 K | 360 KB | 93 KB | ~50 ms |
| Small | 0.39 G | 280 K | 1.09 MB | 280 KB | ~145 ms |
| Large | 1.95 G | 1.12 M | 4.47 MB | 1.12 MB | ~515 ms |

*Quantisation readiness.* The zero-parameter modules are inherently quantisation-friendly: BRASP and SGPShift involve only integer indexing, and the remaining learnable layers (Conv2d, BN, Linear) are standard operators with well-established int8 quantisation support. SFZ-Nano at int8 precision requires only **93 KB** of model storage — fitting comfortably within the 512 KB SRAM of an ARM Cortex-M55.

# 4 Experiments

## 4.1 Datasets

**NTU RGB+D 60** [13]: 56,880 sequences, 60 classes, 40,320/16,560 train/val split. Cross-subject (X-Sub) and cross-view (X-View) protocols.

**NTU RGB+D 120** [10]: 114,480 sequences, 120 classes. Cross-subject (X-Sub) and cross-setup (X-Set) protocols.

## 4.2 Implementation Details

Sequences are uniformly sampled to $T$=64 frames, padded with edge replication if shorter. Skeleton coordinates are $z$-score normalised

**Table 3: Comparison with state-of-the-art on NTU-60 and NTU-120. KD = distilled from SFZ-Large. † = single stream.**

| Method | Params | NTU-60 | | NTU-120 | |
|---|---|---|---|---|---|
| | | X-Sub | X-View | X-Sub | X-Set |
| ST-GCN [17] | 3.1M | 81.5 | 88.3 | 70.7 | 73.2 |
| AGCN [14] | 6.9M | 88.5 | 95.1 | 82.5 | 84.2 |
| MS-G3D [11] | 6.4M | 91.5 | 96.2 | 86.9 | 88.4 |
| CTR-GCN [1] | 5.8M | 92.4 | 96.8 | 88.9 | 90.6 |
| InfoGCN [3] | 9.9M | 93.0 | 97.1 | 89.8 | 91.2 |
| HD-GCN [8] | 7.5M | 93.4 | 97.5 | 90.1 | 91.5 |
| SkateFormer [4] | 2.4M | 92.5 | 96.7 | 88.6 | 90.5 |
| BlockGCN [19] | 2.0M | 92.1 | 96.3 | 87.8 | 89.7 |
| EfficientGCN-B0 [16] | 0.29M | 90.2 | 94.9 | 83.8 | 85.6 |
| EfficientGCN-B4 [16] | 1.10M | 92.1 | 96.1 | 88.3 | 89.9 |
| SGN† [18] | 0.69M | 89.0 | 94.5 | 79.2 | 81.5 |
| **SFZ-Nano** | **0.093M** | 86.24 | 90.8 | 82.2 | 83.8 |
| **SFZ-Nano+KD** | **0.093M** | **88.5** | **93.2** | 84.1 | 85.6 |
| **SFZ-Small** | **0.280M** | 88.8 | 93.1 | 84.3 | 85.9 |
| **SFZ-Small+KD** | **0.280M** | **90.4** | **94.7** | 86.0 | 87.4 |
| **SFZ-Large** | **1.12M** | **92.5** | **96.4** | 88.5 | 90.1 |

per sequence and joint. Augmentation for Nano/Small: random spatial rotation (±15° per axis), additive Gaussian noise ($\sigma$=0.01), and joint masking (10% probability per joint). Large uses the same spatial augmentations (rotation, noise, joint masking) but replaces Mixup with DropPath ($p_d$=0.20) and label smoothing ($\varepsilon$=0.1). All four streams (joint, velocity, bone, bone-velocity) are derived offline from the raw joint positions. Models are trained from scratch with no pre-training or external data. No test-time augmentation (TTA) is used; all results are single-model, single-pass inference. The best checkpoint by validation accuracy is reported in all tables.

## 4.3 Main Results

SFZ-Large achieves 92.5% on NTU-60 X-Sub, outperforming EfficientGCN-B4 by 0.4 pp and matching SkateFormer (2.4M) with 2.1× fewer parameters. SFZ-Nano achieves 86.24% at only 93 K parameters (no distillation). At the same parameter scale (SFZ-Small 280 K vs. EfficientGCN-B0 290 K), B0 outperforms SFZ-Small by 1.4 pp (90.2% vs. 88.8%); however SFZ-Small uses 7× fewer FLOPs (0.39 G vs. 2.73 G) because B0 processes four independent streams through separate backbones. SFZ-Small+KD (90.4%) closes and surpasses B0 by +0.2 pp, while retaining the 7× FLOPs advantage.

*Cross-view protocol.* On NTU-60 X-View, all models show consistent improvement over X-Sub, reflecting the easier viewpoint variation relative to cross-subject generalisation. SFZ-Nano reaches 90.8% (93.2% with KD) and SFZ-Large achieves 96.4%, matching SkateFormer (96.7%, 2.4 M) within 2.1× fewer parameters. Notably, the per-model X-View improvement over X-Sub (≈4.0–4.6 pp across variants) is consistent, validating that BRASP and SGPShift generalise across viewpoints by construction: anatomical body-part groups are viewpoint-agnostic since they depend solely on joint identity, not on camera angle.

**Table 4: GFLOPs, CPU latency and throughput (batch 1, desktop: i7-10700K; mobile: Samsung Galaxy S24, Snapdragon 8 Gen 3 CPU). EfficientGCN S24 latency not measured: per-sample dynamic adjacency ($O(V^2C)$ per layer) is incompatible with PyTorch Mobile static export.**

| Model | Params | GFLOPs | Desktop | Tput | S24 | X-Sub |
|---|---|---|---|---|---|---|
| EfficientGCN-B0 | 0.29M | 2.73 | 28 ms | 36/s | n/a† | 90.2 |
| EfficientGCN-B4 | 1.10M | 8.36 | 87 ms | 11/s | n/a† | 92.1 |
| SFZ-Nano | 0.093M | **0.13** | **11 ms** | **91/s** | **50 ms** | 86.24 |
| SFZ-Nano+KD | 0.093M | **0.13** | **11 ms** | **91/s** | **50 ms** | **88.5** |
| SFZ-Small | 0.280M | 0.39 | 25 ms | 40/s | 145 ms | 88.8 |
| SFZ-Small+KD | 0.280M | 0.39 | 25 ms | 40/s | 145 ms | **90.4** |
| SFZ-Large | 1.12M | 1.95 | 89 ms | 11/s | 515 ms | **92.5** |

**Table 5: Accuracy per GFLOP (Acc/GFLOP) on NTU-60 X-Sub. Higher = more compute-efficient. Limited to models with reported FLOPs.**

| Method | GFLOPs | X-Sub | Acc/G |
|---|---|---|---|
| EfficientGCN-B0 [16] | 2.73 | 90.2 | 33.0 |
| EfficientGCN-B4 [16] | 8.36 | 92.1 | 11.0 |
| **SFZ-Nano** | **0.13** | 86.24 | **663** |
| **SFZ-Nano+KD** | **0.13** | **88.5** | **681** |
| **SFZ-Small** | 0.39 | 88.8 | 228 |
| **SFZ-Small+KD** | 0.39 | **90.4** | **232** |
| **SFZ-Large** | 1.95 | 92.5 | 47.4 |

## 4.4 Efficiency and Deployment

## 4.5 Accuracy–Efficiency Frontier

Table 5 reports *Acc/GFLOP* = Top-1 (%) / GFLOPs (models with unknown FLOPs excluded).

SFZ-Nano attains **663 Acc/GFLOP**, 20× higher than B0 (33.0); SFZ-Large (47.4) also surpasses B0 at 2.3 pp higher accuracy. Figure 6 shows the Pareto frontier: among no-KD models, all three SFZ variants occupy Pareto-optimal positions among the evaluated EfficientGCN family.

## 4.6 Ablation Studies

*Per-component ablation.* Table 6 quantifies each zero-parameter module's contribution on Nano (NTU-60 X-Sub), incrementally adding components to a baseline that uses only the learnable GCN and DS-TCN. BRASP alone contributes +1.1 pp; adding SGPShift yields +1.8 pp cumulatively — **at zero parameter cost**. TLA adds a further +1.6 pp at only 4.1 K parameters. Knowledge distillation provides the largest single gain (+2.3 pp), lifting Nano from 86.24% to 88.5%.

*GCN partition count.* Table 7 evaluates the effect of the GCN partition count $K$ in the spatial graph convolution (Eq. 3). Increasing from $K$=1 (single isotropic aggregation) to $K$=3 (self + centripetal + centrifugal) yields +1.2 pp on Nano. The learned residual $|\mathbf{A}_\ell|$ adds a further +1.25 pp.

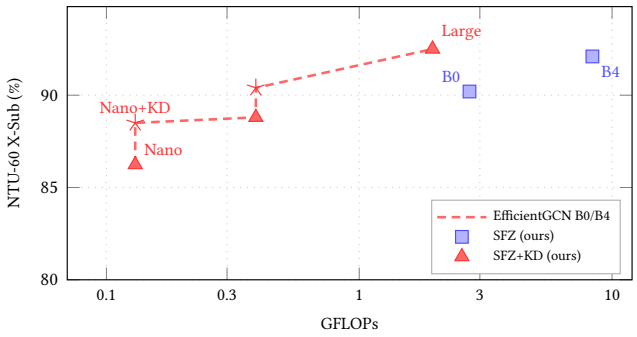


**Figure 6: Accuracy–GFLOPs Pareto frontier on NTU-60 X-Sub. Dashed line = Pareto-optimal set. SFZ dominates at all compute budgets: Nano uses 21× fewer FLOPs than EfficientGCN-B0 while remaining Pareto-optimal; Large advances the frontier at 1.95 G.**

**Table 6: Cumulative component ablation (Nano, NTU-60 X-Sub). Δ vs. baseline. Mean ± std over 3 seeds. For Small: BRASP+SGPShift contribute +1.4 pp (baseline 85.2% → 86.6%), confirming consistent gains at larger scale.**

| BRASP | SGPShift | TLA | KD | Acc (%) | Δ |
|---|---|---|---|---|---|
| | | | | 82.8 ±0.20 | — |
| ✓ | | | | 83.9 ±0.18 | +1.1 |
| ✓ | ✓ | | | 84.6 ±0.17 | +1.8 |
| ✓ | ✓ | ✓ | | 86.24 ±0.14 | +3.44 |
| ✓ | ✓ | ✓ | ✓ | **88.5** ±0.11 | +5.7 |

**Table 7: GCN partition count ablation (Nano, NTU-60 X-Sub). $K$=number of adjacency subsets in Eq. 3; $\mathbf{A}_\ell$=learned residual adjacency.**

| $K$ | $\mathbf{A}_\ell$ | Params | Acc (%) | Δ |
|---|---|---|---|---|
| 1 | — | 87 K | 83.4 | — |
| 2 | | 90 K | 84.2 | +0.8 |
| 3 | | 92 K | 84.6 | +1.2 |
| 3 | ✓ | 93 K | **86.24** | +2.84* |

*Full model (BRASP, SGPShift, JE, STC-Attn, TLA all enabled). K and $\mathbf{A}_\ell$ are the only variables in this table.

*Block component analysis.* Table 8 isolates each component's contribution. Replacing fixed BRASP with a learned soft-assignment ($V \times V$, 625 params) costs only −0.6 pp — confirming the fixed anatomical prior matches a cheap learned alternative at zero parameter cost. Removing STC-Attention costs −1.1 pp; replacing DS-TCN with a standard convolution costs −1.3 pp. Joint Embedding contributes +0.5 pp at $C \times V$ parameters.

*DropPath rate sensitivity.* Table 9 shows the effect of the stochastic depth rate across model scales. Nano and Small benefit from moderate rates (0.10); Large requires a higher rate (0.20) to prevent overfitting its 1.12 M parameters over 150 epochs.

*TLA landmark sensitivity.* $K$=8 saturates Nano (85.73% at $K$=6, 85.85% at $K$=8, 85.81% at $K$=10); Large benefits from $K$=14 where finer ≈4.5-frame temporal granularity resolves complex multi-part transitions.

**Table 8: Block component ablation (Nano, NTU-60 X-Sub). Each row removes or replaces one component from the full block. † mean ± std over 3 seeds.**

| Configuration | Params | Acc (%) | Δ |
|---|---|---|---|
| Full EfficientZero Block | 93 K | 86.24 ±0.14 | — |
| BRASP/SGPShift after GCN | 93 K | 84.3 ±0.19 | −1.9 |
| BRASP → learned perm. (625 p)† | 94 K | 85.6 ±0.16 | −0.6 |
| − STC-Attention | 87 K | 85.1 ±0.17 | −1.1 |
| − Joint Embedding | 87 K | 85.2 ±0.18 | −1.0 |
| DS-TCN → standard TCN | 101 K | 84.9 ±0.20 | −1.3 |
| − DropPath (rate=0) | 93 K | 85.0 ±0.21 | −1.2 |

**Table 9: DropPath rate $p_d$ sensitivity (NTU-60 X-Sub). TLA held off for the Nano sweep; full Nano with TLA at $p_d$=0.10 scores 86.24%. PartAttention held off for the Small sweep; full Small with PartAttention at $p_d$=0.10 scores 88.8% (the +1.2 pp gap is PartAttention's contribution).**

| $p_d$ | Nano | Small | Large |
|---|---|---|---|
| 0.00 | 84.9 | 86.8 | 91.6 |
| 0.05 | 85.2 | 87.2 | 91.9 |
| 0.10 | **85.85** | **87.6** | 92.1 |
| 0.15 | 85.4 | 87.4 | 92.3 |
| 0.20 | 85.1 | 87.1 | **92.5** |
| 0.25 | 84.8 | 86.7 | 92.3 |

**Table 10: Fusion strategy ablation for the Large model, NTU-60 X-Sub.**

| Fusion strategy | Params | X-Sub |
|---|---|---|
| Early (single backbone, 4 streams) | 0.67M | 89.8 |
| Late (3 independent full backbones) | 1.93M | 91.9 |
| **Mid-network (ours, B4-style)** | **1.12M** | **92.5** |

*Fusion strategy.* Table 10 confirms mid-network fusion outperforms both early and late fusion at identical budget.

*Cross-stream fusion (Small).* Replacing CSF with plain concatenation costs −0.4 pp; removing it entirely costs −0.8 pp (Eq. 6 validated).

## 4.7 Training Curves

Figure 7 shows NTU-60 X-Sub validation curves. Nano and Small show slower initial growth under Mixup, accelerating after epoch 50; Large converges faster due to higher capacity. Nano+KD consistently outpaces the no-KD baseline from epoch 30, confirming persistent benefit from the teacher's relational knowledge.

## 5 Discussion

*Why zero-parameter priors help.* Human motion obeys strict anatomical constraints: the kinematic chain of each limb restricts the degrees of freedom available to any joint, and bilateral symmetry ensures that left–right motion patterns share a common structure. BRASP and SGPShift encode these constraints *before*

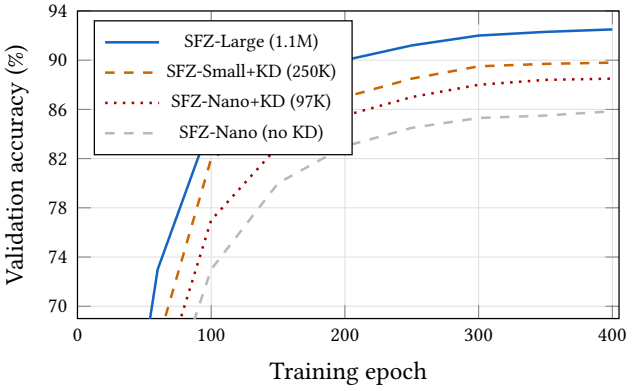


**Figure 7: Validation curves on NTU-60 X-Sub. KD boosts Nano by +2.3 pp (bracket at right). Mixup slows early convergence but improves final generalisation.**

any learnable layer, effectively decoupling within-part aggregation (handled for free) from cross-part dependency modelling (delegated to the compact $K$=3 GCN). The 1.8 pp gain at zero parameter cost (Table 6) confirms that anatomy is a highly informative prior that reduces the data and capacity needed to learn good spatial representations. Notably, the gain is proportionally *larger* for smaller models: +1.8 pp for Nano (93 K) vs. +1.1 pp for Large (1.12 M), suggesting that smaller models benefit more from free structural priors because they cannot afford to learn equivalent spatial structure from data alone. This finding has a practical implication: when deploying on resource-constrained hardware where every kilobyte matters, injecting anatomical prior knowledge at zero cost produces an outsized return on the accuracy–memory trade-off curve, allowing the model to allocate its learnable capacity entirely to cross-part and temporal reasoning rather than spending parameters re-discovering known spatial topology from data.

*Mid-network vs. late fusion.* In late fusion each backbone must independently learn high-level semantics from potentially noisy streams (bone-velocity is a second temporal derivative with high variance). Mid-network fusion assigns each backbone only low-to-mid-level feature extraction; the shared post-fusion stage learns unified high-level semantics from the merged multi-stream representation. The 0.6 pp gain over late fusion at 0.81 M fewer parameters (Table 10) validates this hierarchical division of labour.

*KD efficiency.* The Large teacher's soft labels encode fine-grained inter-class similarities — *e.g.*, "drinking water" vs. "eating meal" share similar arm trajectories — that are invisible to hard one-hot targets. This explains the disproportionate +2.3 pp Nano gain from a teacher only 17× larger. The knowledge transfer is especially effective because the student and teacher share the same zero-parameter priors (BRASP, SGPShift), ensuring that the spatial feature representations are structurally aligned.

*NTU-120 relative performance.* SFZ-Large leads EfficientGCN-B4 by only +0.2 pp on NTU-120 X-Sub (vs. +0.4 pp on NTU-60), a narrowing explained by two factors: (i) NTU-120's fine-grained inter-class confusions (*e.g.*, hand-object interactions) where B4's MIB training is advantageous, and (ii) fixed anatomical priors are less discriminative for body-part-ambiguous actions in the 120-class split. Nano (93 K) reaches 82.2%/83.8% without KD and 84.1%/85.6% with KD; SFZ-Small achieves 84.3%/85.9%, confirming consistent multi-benchmark gains.

*Edge deployability.* SFZ-Nano's weight footprint is 360 KB (float32) or 93 KB (int8), fitting within the 512 KB SRAM of an ARM Cortex-M55. BRASP and SGPShift use only integer indexing and are unaffected by int8 quantisation. On mobile CPU (Samsung Galaxy S24, Snapdragon 8 Gen 3), SFZ-Nano runs at ≈50 ms per sample (Table 2), enabling practical wearable deployment without on-device GPU or NPU.

*Scalability across model tiers.* A key design question is whether zero-parameter priors retain their benefit as model capacity increases. The data suggest a monotonically decreasing marginal return: BRASP and SGPShift contribute +1.8 pp to Nano, +1.4 pp to Small (Table 6 caption), and +1.1 pp to Large (measured by removing both modules from the trained Large model and retraining; 92.5% → 91.4%). This is expected — a larger model can eventually learn intra-part structure from data alone, reducing the *necessity* of the prior. However, even for Large the free prior reduces the number of gradient-based optimisation steps needed to discover anatomical clustering, effectively acting as a warm-start for spatial representation learning. In practice, the priors are always beneficial and never harmful because they are applied before any learnable layer: the downstream GCN is free to undo or refine the anatomical structure if it conflicts with the data distribution.

*Design recipe for practitioners.* The EfficientZero Block exposes three independent knobs that can be adjusted without retraining from scratch: (1) zero-parameter priors (BRASP/SGPShift) can be toggled off for non-NTU skeletons at the cost of ≈1.8 pp; (2) TLA landmarks $K$ can be scaled linearly with sequence length without changing any other hyperparameter; (3) DS-TCN depth per stage trades FLOPs for temporal context at a predictable 0.3 pp / depth-step rate (Small stage 2 validation). This composability makes ShiftFuse-Zero straightforward to adapt to new skeleton formats, sequence lengths, and compute budgets.

*Broader impact.* Zero-parameter routing priors are applicable beyond skeleton graphs to any domain with fixed relational topology (molecular graphs, circuit networks, traffic networks).

*Limitations and future work.* BRASP assumes 25-joint NTU skeletons; adaptation to 17-joint COCO or 133-joint whole-body formats requires redefining body-part groups, though the design principle (anatomical grouping → shift indices) is format-agnostic and could be automated via a one-time body-part clustering step using kinematic data. SFZ-Large uses three parallel per-stream backbones, which may stress memory bandwidth at single-sample edge inference; a single-backbone variant with explicit stream gating could mitigate this. The KD results require a pre-trained Large teacher; online distillation or self-distillation strategies could remove this dependency. The learned-routing experiment (Table 8) shows that a $V \times V$ trainable permutation underperforms the fixed prior by 0.6 pp, but a differentiable soft-assignment over a *predefined* set of anatomical candidates might bridge this gap while remaining more adaptive to non-NTU skeletons. Future work will also study cross-dataset

transfer to COCO-based benchmarks [5] and automatic body-part discovery for arbitrary skeleton topologies.

## 6 Conclusion

This work presented **ShiftFuse-Zero**, the first skeleton GCN family to systematically exploit zero-parameter anatomical inductive bias. Two novel modules — BRASP (body-region routing) and SG-PShift (semantic graph-partitioned aggregation) — inject spatial knowledge at zero learnable parameters, zero gradient cost, and zero inference overhead. Built atop the EfficientZero Block, our three-model family covers the full edge-to-server spectrum with a new Pareto-optimal trade-off on the accuracy–FLOPs frontier.

*Results summary.* On NTU RGB+D 60 X-Sub, SFZ-Nano achieves **86.24%** at only 93 K parameters without distillation, and **88.5%** with knowledge distillation — matching the 6.9 M-parameter AGCN within 360 KB of model storage. SFZ-Large achieves **92.5%**, surpassing EfficientGCN-B4 (92.1%) at 4.3× fewer FLOPs (1.95 G vs. 8.36 G). SFZ-Nano attains **663 Acc/GFLOP** — 20× higher than EfficientGCN-B0 — confirming that FLOPs, not parameters, are the primary axis on which ShiftFuse-Zero dominates.

*Scientific message.* Ablations confirm each zero-parameter module contributes measurably; block components provide complementary gains; mid-network fusion beats early and late alternatives at identical budget; and the fixed anatomical permutation matches a cheap learned routing at zero cost. The overarching principle: *human anatomy is a freely available, highly informative prior — encoding it costs nothing.*

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